

**TO:** Cook County Board of Commissioners

**FROM:** Haedodam Kim

**DATE:** May 3, 2025

**SUBJECT:** Reducing Geographic Disparities in Trauma Care Access in Cook County

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## **EXECUTIVE SUMMARY**

There are severe regional disparities in access to trauma care across Cook County. While downtown and northern areas benefit from strong proximity to trauma centers and rapid emergency response, the southern region remains entirely without trauma care infrastructure. As a result, patients must often be transported more than 20 minutes to reach a hospital following a trauma incident, which directly impacts survival rates.

This report diagnoses the issue using a five-step sequential model and evaluates four policy alternatives through a weighted framework: Effectiveness (40%), Feasibility (20%), Cost (20%), and Time to Implement (20%). The four policy alternatives examined are: Establishing a Trauma Center in the Southland, Decentralized Trauma Response Network, EMS Resource Reallocation and Dispatch Protocol Adjustment, and Implementation of Air Medical Transport Systems.

Among the proposed options, the most balanced and realistic is the establishment of a Decentralized Trauma Response Network. This model involves installing satellite trauma units in the Southland in coordination with major trauma centers. These units provide essential services such as initial stabilization and pre-transfer emergency care. Although they are not full-scale trauma centers, their ability to deliver timely intervention during the critical window can significantly improve survival outcomes. According to a BOTEK, the first-year setup and operating cost for a single satellite unit is estimated at \$3.75–4.95 million, making it far more cost-effective than building a new trauma center, which may cost \$20–50 million annually. Implementation can also be achieved within 6 to 12 months, and the model can be further integrated with future interventions such as EMS reallocation or air medical transport systems.

## **PROBLEM DEFINITION** *(See Appendix 1 and 2)*

Trauma care access in Cook County varies drastically by region. While downtown, northern, and western areas benefit from a dense network of trauma centers with transport times of 5 to 10 minutes, the South Side and Southland, where gun violence and severe trauma are most

prevalent, have long been isolated from such infrastructure. Since the closure of St. James Olympia Fields in 2008, the Southland has had no trauma center, forcing patients to be transported more than 20 minutes to reach appropriate care. A 2015 study by the Illinois Department of Public Health found that trauma patients injured more than five miles from a trauma center face a 23 percent higher risk of death. This reflects a structural failure to deliver immediate and specialized treatment and presents a serious public health crisis. Cook County and community advocates have pushed for the establishment of a trauma center in the Southland. However, progress has stalled due to projected annual operating costs exceeding \$20 million. If the County aims to build an equitable and resilient public health system, policy and fiscal intervention to address these geographic disparities must no longer be postponed.

### **PROBLEM DIAGNOSIS MODEL** *(See Appendix 3)*

This issue can be structurally diagnosed through the following five-step sequential process.

#### **1. Trauma Incident Occurrence**

Severe trauma incidents such as gun violence, motor vehicle collisions, and industrial accidents occur disproportionately in the southern and outlying areas of Cook County. These communities are characterized by high rates of poverty and unemployment, lower levels of educational attainment, and a demographic concentration of trauma-vulnerable populations, particularly young adult males. As a result, these areas face structurally elevated risks of life-threatening injuries.

#### **2. EMS Dispatch and On-Scene Arrival**

When a trauma event occurs, the emergency medical system (EMS) is activated through a 911 call. However, southern Cook County suffers from limited ambulance availability, staffing shortages, and stricter dispatch thresholds compared to central regions. EMS response times in these areas often range from 15 to 30 minutes, well beyond the nationally recommended standard of eight minutes. This delay represents a critical bottleneck, frequently resulting in missed opportunities during the golden hour when rapid treatment is most likely to save lives.

#### **3. Hospital Transport Decision**

Due to the absence of trauma centers in the Southland, EMS personnel must transport patients 10 to 15 miles or more to reach the nearest qualifying facility. This situation effectively eliminates hospital choice for patients and imposes a heavy burden on trauma centers in other parts of the county. These facilities are often overextended, leading to long wait times or secondary transfers, which introduce further delays in care.

#### **4. Patient Transport**

Long-distance travel to trauma centers introduces additional time loss. Transport durations commonly range from 17 to 30 minutes depending on traffic and road conditions. During this period, patient conditions may deteriorate, and in some cases, fatalities occur in transit. Upon arrival, overcrowded hospitals may be unable to provide immediate care, causing further delays or necessitating transfers to alternative facilities.

#### **5. Hospital Arrival and Trauma Care Initiation**

By the time patients reach a trauma center, care may begin after the critical intervention window has passed. According to a 2015 study by the Illinois Department of Public Health, trauma victims injured more than five miles from a trauma center face a 23% higher risk of death. This stark figure underscores how geographic and infrastructure disparities translate directly into time-based inequities in survival outcomes.

The most effective stages for policy intervention are threefold. First, EMS dispatch and on-scene arrival is a critical point that determines initial response time and reflects underlying disparities in resource distribution. Second, the hospital transport decision stage poses a structural challenge due to the absence of nearby trauma centers, severely limiting patients' choice of destination. Third, the patient transport stage directly impacts survival rates, as long-distance travel introduces significant delays in time-critical emergencies.

### **POLICY ALTERNATIVES**

#### **1. Establishing a Trauma Center in the Southland**

This alternative proposes building a new trauma center in the Southland, where no such facility currently exists. It would significantly improve access to trauma care, expand patient choice, relieve pressure on downtown trauma centers, and reduce emergency transport times, ultimately lowering mortality rates. However, construction may cost tens of millions of dollars, and annual operating expenses could exceed \$20 million. Maintaining 24/7 operations and securing specialized trauma personnel would impose substantial financial and administrative burdens on Cook County.

#### **2. Decentralized Trauma Response Network**

This option involves creating satellite trauma units in the Southland in coordination with existing major trauma centers. These units would provide initial stabilization, triage, and basic emergency care before transferring patients to higher-level centers based on injury severity. It

requires lower capital investment and can be implemented more quickly than building a full trauma center. However, because satellite units typically lack surgical capabilities, severe trauma cases would still require transfer.

### **3. EMS Resource Reallocation and Dispatch Protocol Adjustment**

This alternative proposes reallocating EMS personnel and vehicles to high-risk areas like the Southland and adjusting 911 dispatch criteria based on local trauma risk. It is a low-cost and immediately implementable strategy. The main drawbacks include possible reduced EMS coverage in already well-served areas and the fact that transport times will remain long if trauma centers remain distant.

### **4. Implementation of Air Medical Transport Systems**

This alternative introduces helicopter-based emergency transport to reduce the time to care in areas far from trauma centers. Air transport can significantly shorten prehospital time, especially in trauma deserts. However, the approach faces high operating costs, weather and night flight limitations, noise concerns, and potential community resistance.

## **DECISION MATRIX** *(See Appendix 4)*

To evaluate the proposed policy alternatives, the following four criteria are applied. All alternatives are scored based on weighted criteria.

**a. Effectiveness(40%):** This criterion evaluates how well each policy alternative can reduce geographic disparities in trauma care access. Key factors include the ability to shorten emergency transport times and distances, lower mortality rates, and expand coverage in high-risk areas.

**b. Feasibility(20%):** Feasibility assesses whether each alternative can realistically be implemented from political, administrative, and institutional perspectives. This includes considerations such as funding approval, intergovernmental cooperation, ease of partnership with hospitals, and regulatory compliance. Options requiring complex coordination or facing legal barriers may score lower.

**c. Cost(20%):** This criterion compares the financial burden of each alternative, including both initial capital investment and annual operating expenses.

**d. Time to Implement(20%):** This measures the estimated time required for each alternative to be implemented and for its benefits to reach residents. While feasibility asks whether a policy can be implemented, this criterion asks when its impact will be realized.

## POLICY EVALUATION

### 1. Establishing a Trauma Center in the Southland (Weighted Score: 14 / 20)

- a. Effectiveness (5 Points): Establishing a new trauma center in the medically underserved Southland region would significantly reduce transport distance and emergency response time, directly contributing to improved survival rates. It would also alleviate the overload at downtown trauma centers, enhancing the overall efficiency of the trauma care system.
- b. Feasibility (2 Points): Constructing a trauma center requires a complex and lengthy administrative process, including state-level approvals, certification from the Department of Public Health, land acquisition, building permits, and staff recruitment. In addition, the facility must meet higher standards than general hospitals, including 24/7 trauma surgeons, emergency nursing teams, and a fully equipped surgical infrastructure.
- c. Cost (1 Point): According to comparable cases in the United States, the initial construction costs for a trauma center can reach tens of millions of dollars. Annual operating expenses are estimated at \$20 to \$50 million, covering advanced medical equipment, specialized personnel, ICU maintenance, and operating room functions.
- d. Time to Implement (1 Point): The full implementation of a new trauma center typically takes 2 to 5 years. This includes the time required for regulatory approvals, architectural planning and construction, equipment procurement, staff onboarding, and official certification. As a result, residents may face a considerable delay before experiencing improved access to trauma care.

### 2. Decentralized Trauma Response Network (Weighted Score: 20 / 20)

- a. Effectiveness (4 Points): Although satellite trauma units are not full-scale trauma centers, they can provide initial triage, patient stabilization, and pre-transfer emergency care. These functions help reduce transport distances and time to treatment, contributing to improved survival outcomes. In addition, satellite units allow for early severity assessment and appropriate allocation of patients to higher-level care facilities.
- b. Feasibility (4 Points): This model typically operates through formal agreements with existing trauma centers, making it administratively simpler than building new facilities.
- c. Cost (4 Points): While some initial investment is required for facility expansion, equipment, and staffing, the costs are significantly lower than those of constructing a

new trauma center. Operating expenses are generally manageable, estimated at several million dollars per year depending on scale and scope.

- d. Time to Implement (4 Points): Satellite trauma units can often be established within 6 to 12 months, particularly when renovations or expansions are made within existing hospital buildings. In facilities with available infrastructure and administrative readiness, the timeline may be even shorter.

### **3. EMS Resource Reallocation and Dispatch Protocol Adjustment (Weighted Score: 18 / 20)**

- a. Effectiveness (2 Points): Reducing EMS response times can have a direct impact on patient survival by enabling faster on-scene care. However, this option does not address the root causes of infrastructure disparities.
- b. Feasibility (4 Points): Because this strategy relies on reallocating existing resources, it can be implemented without formal hospital approval or extensive regulatory processes. Local governments can enact changes to EMS deployment plans within a short time frame at the administrative level.
- c. Cost (5 Points): The required adjustments primarily involve redistributing personnel and potentially adding a limited number of vehicles or equipment.
- d. Time to Implement (5 Points): With only minor operational guideline changes and personnel reassignments needed, this alternative can be put into effect within just a few weeks.

### **4. Implementation of Air Medical Transport System (Weighted Score: 16 / 20)**

- a. Effectiveness (4 Points): It can significantly reduce emergency response times for residents in remote or underserved areas. This solution is particularly effective in addressing the elevated mortality risk. By bypassing ground traffic and long-distance routes, this option improves patient survival rates in critical situations.
- b. Feasibility (3 Points): The implementation of an air medical system involves compliance with complex requirements, including Federal Aviation Administration safety regulations, designated helipad construction, and protocols for onboard medical staffing. There are also potential community concerns such as noise and flight safety.
- c. Cost (2 Points): Annual operating costs for a single helicopter are estimated at \$2 to \$5 million, including pilot salaries, medical crew staffing, insurance, maintenance, and fuel.
- d. Time to Implement (3 Points): the system could become operational within 6 to 12 months. However, implementation timelines may vary based on the availability of landing infrastructure and the speed of regulatory approval.

## **POLICY RECOMMENDATION**

I recommend the implementation of a Decentralized Trauma Response Network by establishing satellite trauma units in the Southland region of Cook County. These units, in collaboration with major trauma centers, would provide core services such as initial emergency stabilization, triage, and pre-transfer care. Although not full-scale trauma centers, these facilities can enable timely interventions during the critical window following trauma, significantly increasing patient survival rates.

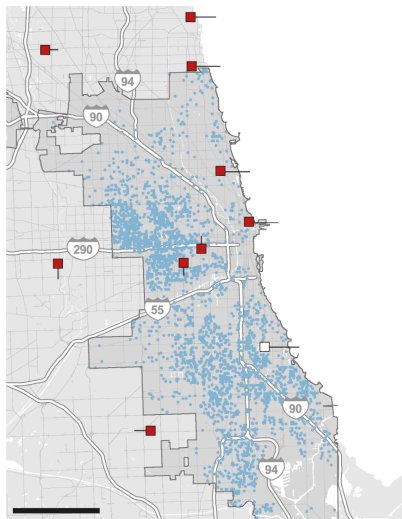
From a policy and administrative standpoint, this alternative is highly feasible and could be implemented within 6 to 12 months. Implementation should proceed through the following steps:

1. The Cook County Board of Commissioners should formally prioritize trauma care equity and adopt a resolution supporting the establishment of satellite trauma units.
2. MOUs should be signed between the County and existing trauma centers to designate appropriate partner hospitals in the Southland.
3. Approval from the Illinois Department of Public Health must be obtained to verify that facility modifications comply with emergency medical service standards.
4. Budget allocation should proceed in parallel. According to BOTECH, the total cost of establishing and operating one satellite trauma unit for one year is estimated at \$3,750,000–\$4,950,000 — including \$800,000 for facility renovations, \$1.2 million for medical equipment, \$1.5 million for annual staffing, and \$500,000 for operational costs. This is far more cost-effective than the \$20–50 million annual cost of a full-scale trauma center.

This option offers a realistic and phased solution to structural disparities by reducing transport times, enabling early stabilization, and expanding emergency service capacity in underserved, high-risk areas. Furthermore, it can be integrated with future interventions such as EMS resource reallocation or air medical transport systems, increasing its policy flexibility and long-term impact.

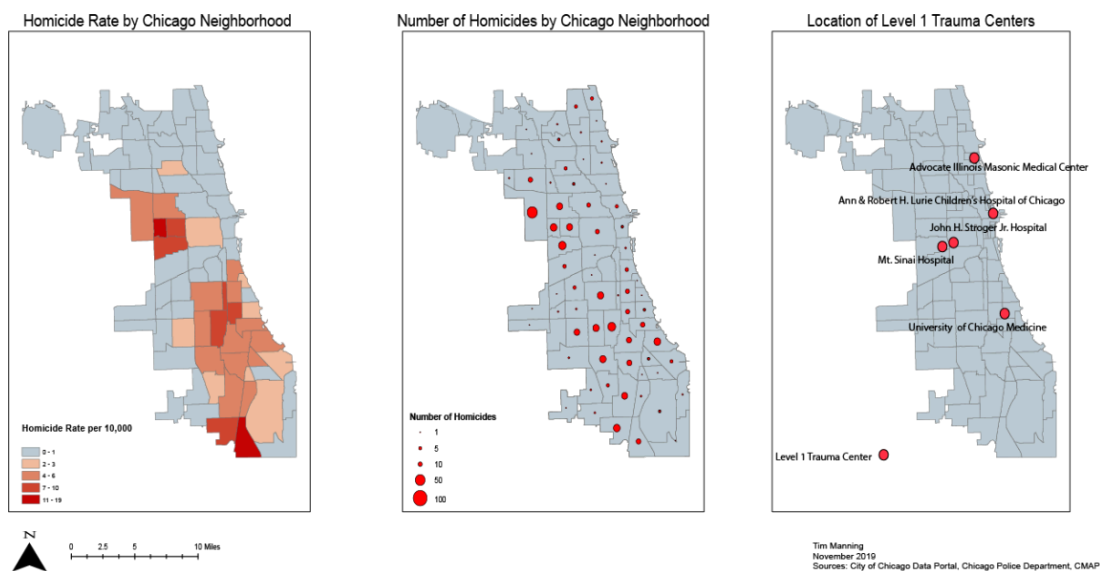
## Appendix

### Appendix 1. Geographic Distribution of Trauma Centers and Incident Density in Cook County

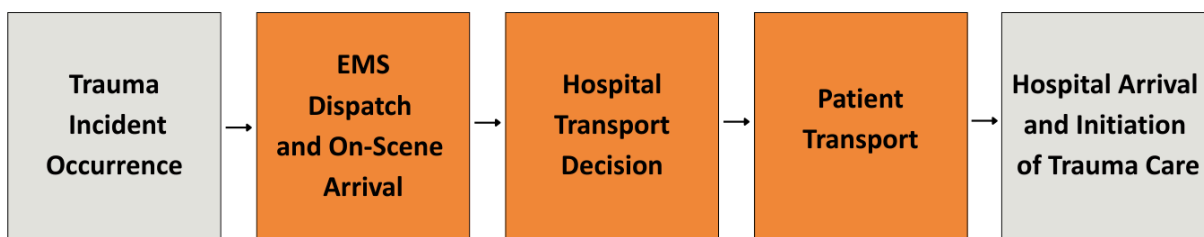


### Appendix 2. Homicides and Trauma Center Locations in Chicago 2018

#### Homicides and Trauma Center Locations in Chicago 2018



### Appendix 3. Problem Diagnosis Model (Intervention Points are highlighted in orange colors)





## Appendix 4. Decision Matrix

|   | Alternatives     | Effectiveness | Feasibility | Cost | Time | Total | Weighted Score |
|---|------------------|---------------|-------------|------|------|-------|----------------|
| 0 | Status Quo       | 1             | 5           | 5    | 5    | 16    | 17             |
| 1 | Trauma Center    | 5             | 2           | 1    | 1    | 9     | 14             |
| 2 | Satellite Units  | 4             | 4           | 4    | 4    | 16    | 20             |
| 3 | EMS Reallocation | 2             | 4           | 5    | 5    | 16    | 18             |
| 4 | Air Transport    | 4             | 3           | 2    | 3    | 12    | 16             |

## Appendix 5. BOTEC

|    | Category                         | Detailed Cost Items                | Estimated Cost        |
|----|----------------------------------|------------------------------------|-----------------------|
| 1  | Facility Renovation              | Emergency room expansion           | 250,000 - 350,000     |
| 2  |                                  | Infrastructure upgrades            | 250,000 - 350,000     |
| 3  |                                  | Sanitary installation & HVAC       | 150,000 - 250,000     |
| 5  | Medical Equipment & Installation | Trauma beds and stretchers         | 350,000 - 450,000     |
| 6  |                                  | Monitoring devices & ventilators   | 450,000 - 550,000     |
| 7  |                                  | Ultrasound/CT and diagnostic tools | 250,000 - 350,000     |
| 9  | Personnel a year                 | Emergency physicians               | 550,000 - 650,000     |
| 10 |                                  | Registered nurses                  | 450,000 - 550,000     |
| 11 |                                  | Paramedics and EMS technicians     | 150,000 - 250,000     |
| 12 |                                  | Administrative and support staff   | 150,000 - 250,000     |
| 13 | Operational Supplies & Utilities | Emergency medical supplies         | 450,000 - 550,000     |
| 14 |                                  | Utilities                          | 150,000 - 250,000     |
| 15 |                                  | Maintenance, insurance, and IT     | 150,000               |
| 18 | Total                            |                                    | 3,750,000 - 4,950,000 |

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Kight, S. (n.d.). *Homicides and Trauma Centers in Chicago* [Map]. Wikimedia Commons.

[https://commons.wikimedia.org/wiki/File:Homicides\\_and\\_Trauma\\_Centers\\_in\\_Chicago.svg](https://commons.wikimedia.org/wiki/File:Homicides_and_Trauma_Centers_in_Chicago.svg)